

Skipping A4 at 2025-11-21 00:00:00 בגלל error: ARIMAGARCH optimization failed: ABNORMAL:
Skipping D1 at 2022-12-23 00:00:00 בגלל error: ARIMAGARCH optimization failed:
ABNORMAL:
Skipping D1 at 2025-05-09 00:00:00 בגלל error: ARIMAGARCH optimization failed:
ABNORMAL:
Skipping D1 at 2025-11-14 00:00:00 בגלל error: ARIMAGARCH optimization failed:
ABNORMAL:

Per-level summaries:

	mean_daily_simple_return	total_simple_return	annualized_simple_return \
A4	0.003630	1.120195	1.485517
D4	0.003463	1.054255	1.382211
D3	0.003688	1.148752	1.514959
D2	0.002675	0.721440	0.924973
D1	0.005739	2.179526	3.116616

	annualized_volatility	sharpe_like	final_cumulative_growth
A4	0.072231	12.663746	2.120195
D4	0.080666	10.819111	2.054255
D3	0.105369	8.820681	2.148752
D2	0.193720	3.479307	1.721440
D1	0.234504	6.167378	3.179526

Combined forecast-aware wavelet summary:

mean_daily_simple_return 0.003773
total_simple_return 1.160529
annualized_simple_return 1.577888
annualized_volatility 0.063273
sharpe_like 15.026027
final_cumulative_growth 2.160529
dtype: float64

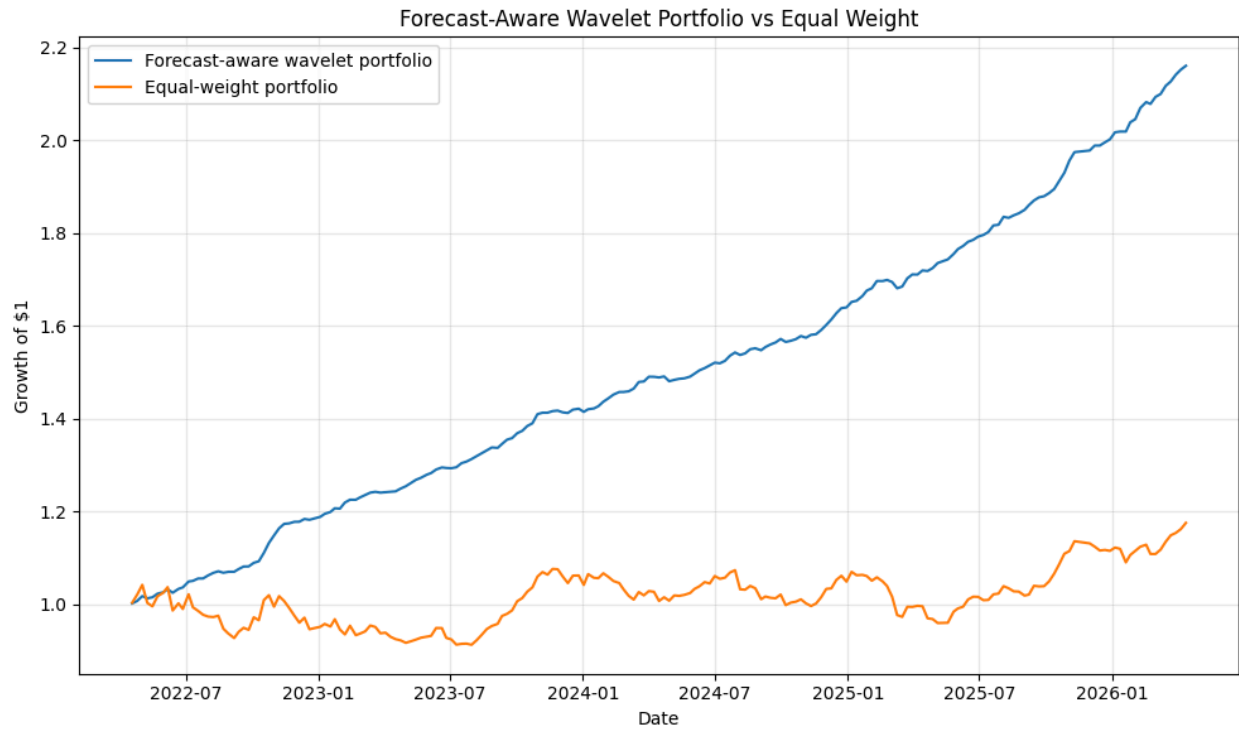
Equal-weight summary:

mean_daily_simple_return 0.000883
total_simple_return 0.175801
annualized_simple_return 0.220278
annualized_volatility 0.215906
sharpe_like 1.030242
final_cumulative_growth 1.175801
dtype: float64

Comparison table:

	ForecastAwareWavelet	EqualWeight
mean_daily_simple_return	0.003773	0.000883

total_simple_return	1.160529	0.175801
annualized_simple_return	1.577888	0.220278
annualized_volatility	0.063273	0.215906
sharpe_like	15.026027	1.030242
final_cumulative_growth	2.160529	1.175801



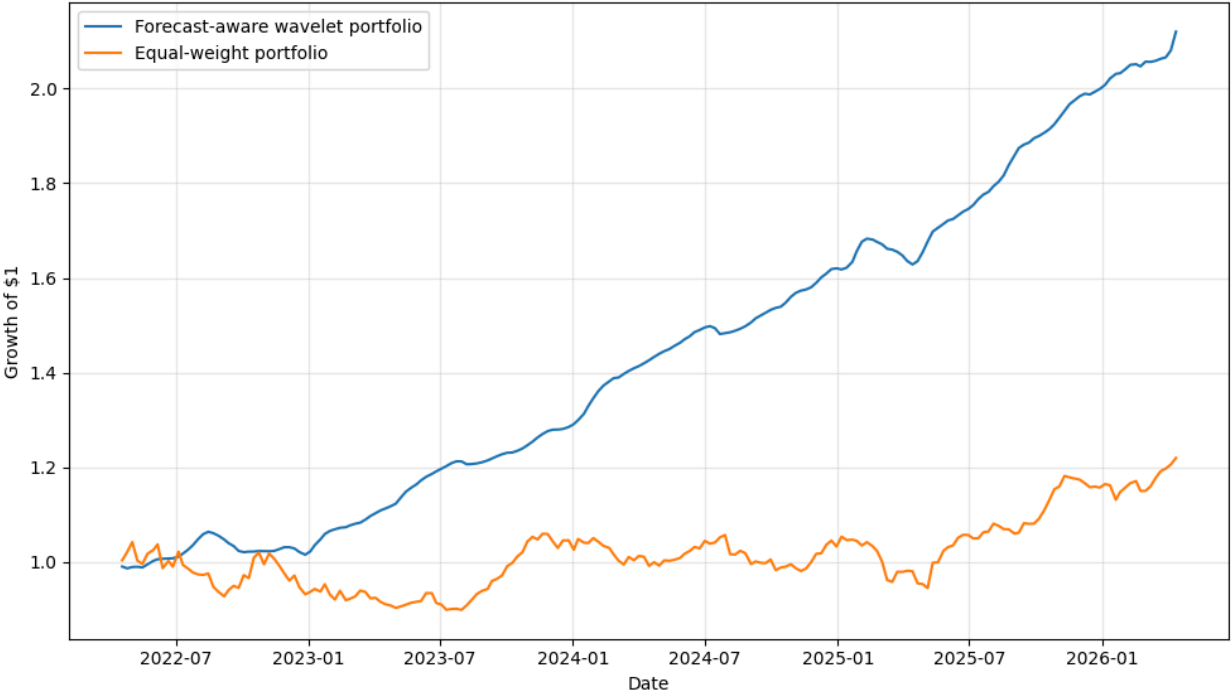
A4-only forecast-aware summary:

mean_daily_simple_return	0.003630
total_simple_return	1.120195
annualized_simple_return	1.485517
annualized_volatility	0.072231
sharpe_like	12.663746
final_cumulative_growth	2.120195
dtype:	float64

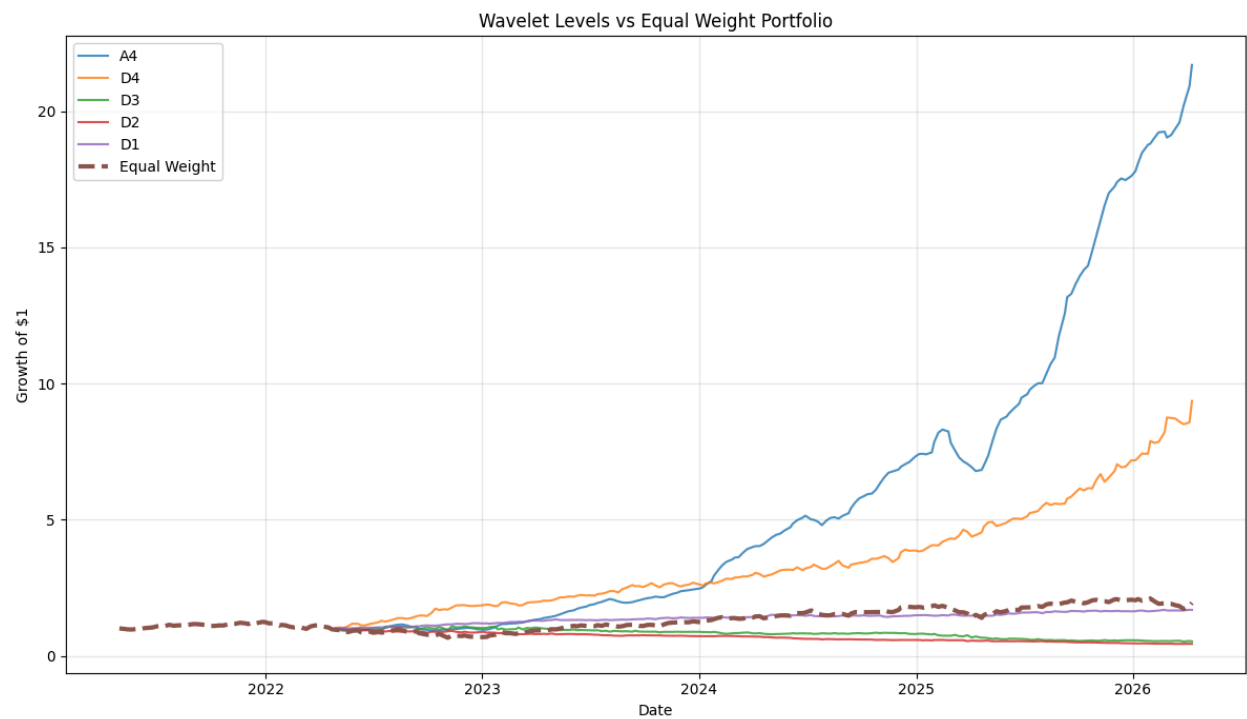
A4-only vs Equal-weight:

	ForecastAwareWavelet	EqualWeight
mean_daily_simple_return	0.003630	0.001055
total_simple_return	1.120195	0.219893
annualized_simple_return	1.485517	0.272279
annualized_volatility	0.072231	0.223745
sharpe_like	12.663746	1.188205
final_cumulative_growth	2.120195	1.219893

A4-Only Forecast-Aware Wavelet Portfolio vs Equal Weight



weekly forecast weekly rebalance reconstructed

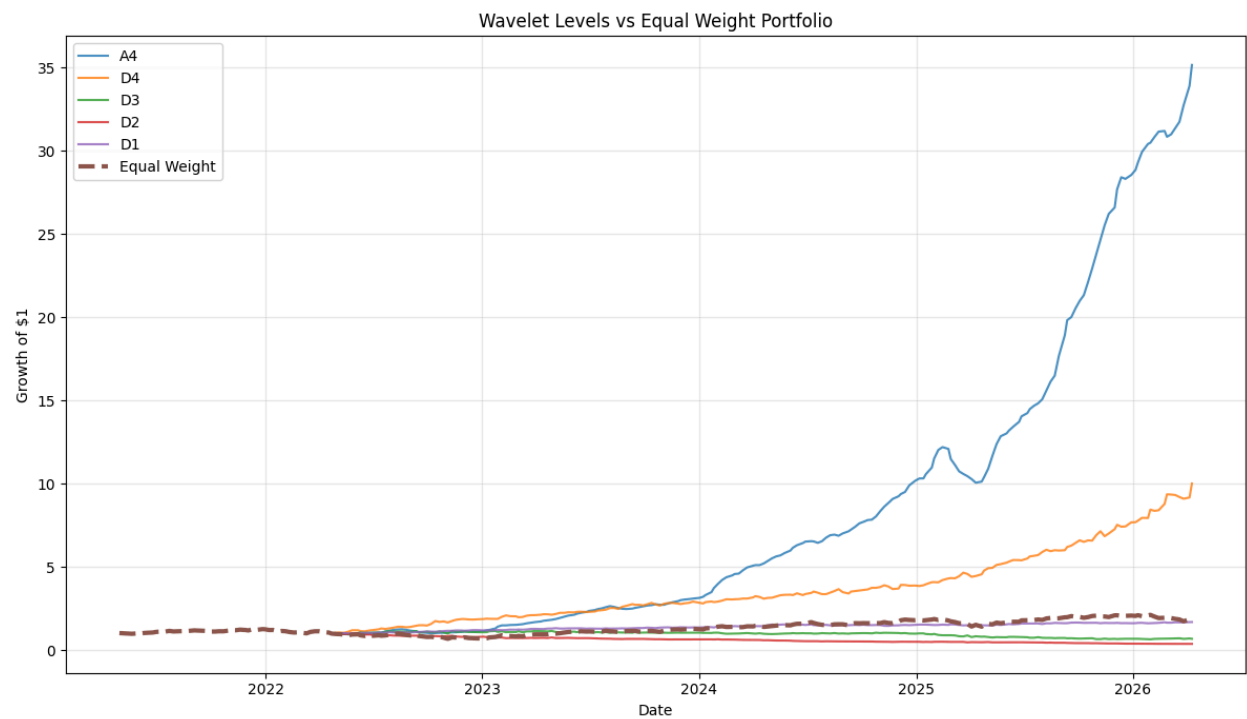


	mean_period_simple_return	total_simple_return \
A4	0.015375	20.694515
D4	0.011171	8.363081
D3	-0.002557	-0.460813
D2	-0.003780	-0.553171
D1	0.002693	0.694083
EqualWeight	0.003169	0.954888

	annualized_simple_return	annualized_volatility	sharpe_like \
A4	1.182612	0.163135	4.900821
D4	0.753993	0.180507	3.217992
D3	-0.144375	0.213900	-0.621700
D2	-0.183205	0.103590	-1.897683
D1	0.146112	0.083631	1.674276
EqualWeight	0.144061	0.245952	0.669943

	final_cumulative_growth
A4	21.694515
D4	9.363081
D3	0.539187
D2	0.446829
D1	1.694083
EqualWeight	1.954888

Daily forecast weekly rebalance reconstructed



	mean_period_simple_return	total_simple_return \
A4	0.017759	34.169125
D4	0.011495	9.008220
D3	-0.001466	-0.325388
D2	-0.004701	-0.632076
D1	0.002641	0.681434
EqualWeight	0.003169	0.954889

	annualized_simple_return	annualized_volatility	sharpe_like \
A4	1.467159	0.159628	5.785014
D4	0.783600	0.180232	3.316472
D3	-0.094148	0.213580	-0.356873
D2	-0.222116	0.110539	-2.211253
D1	0.142377	0.090562	1.516547
EqualWeight	0.144061	0.245952	0.669944

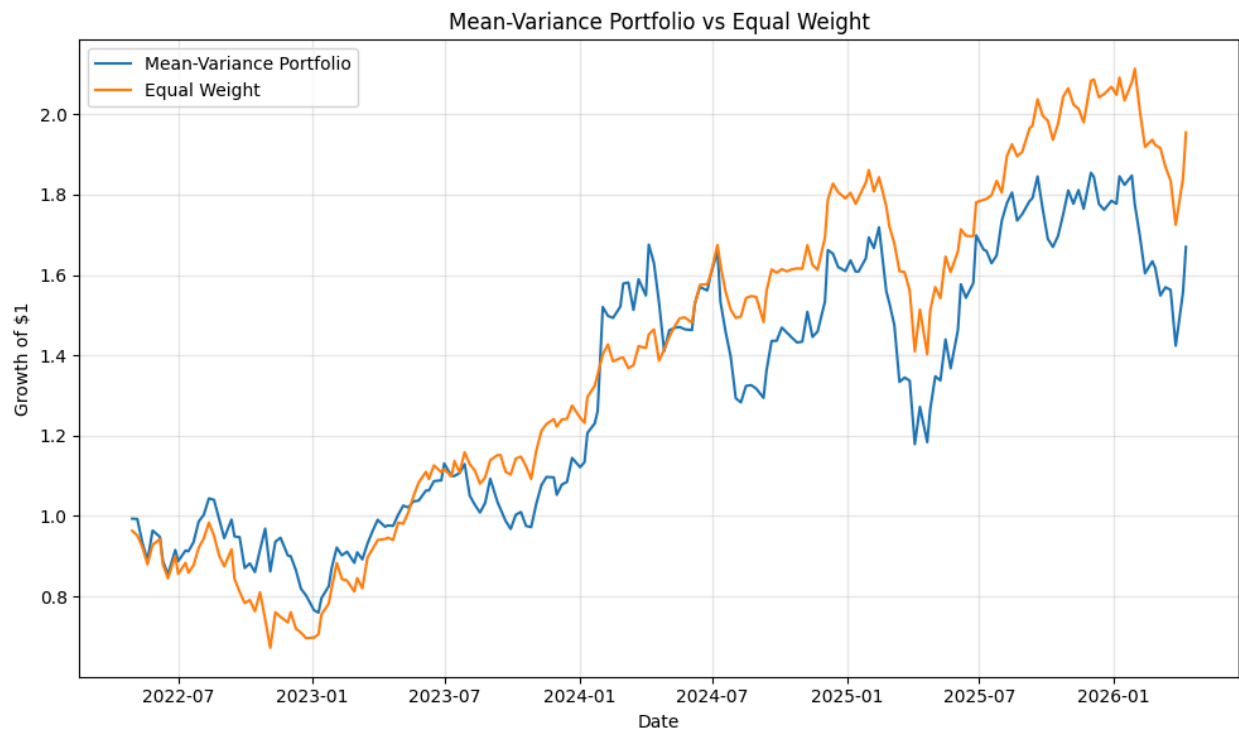
	final_cumulative_growth
A4	35.169125
D4	10.008220
D3	0.674612
D2	0.367924
D1	1.681434
EqualWeight	1.954889

Mean-variance summary:

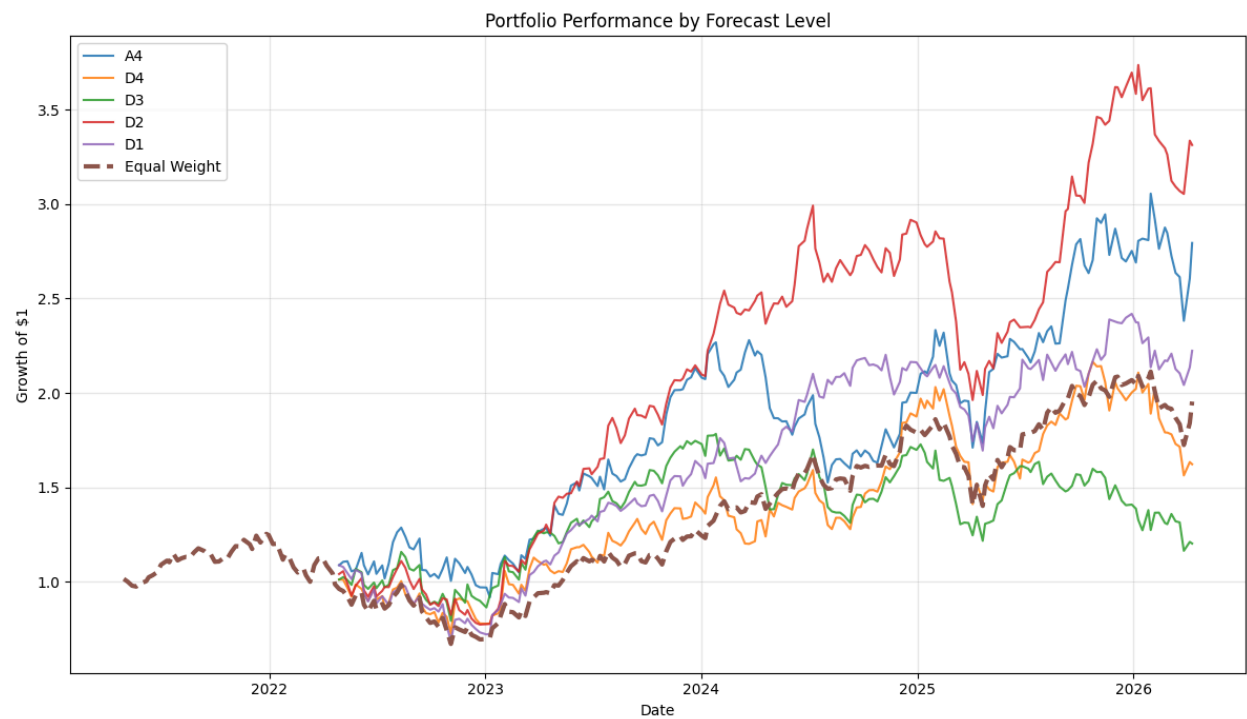
mean_period_simple_return 0.003396
total_simple_return 0.670214
annualized_simple_return 0.137528
annualized_volatility 0.310361
sharpe_like 0.568948
final_cumulative_growth 1.670214
dtype: float64

Equal-weight summary:

mean_period_simple_return 0.003169
total_simple_return 0.954888
annualized_simple_return 0.144061
annualized_volatility 0.245952
sharpe_like 0.669943
final_cumulative_growth 1.954888
dtype: float64



weekly forecast weekly rebalance



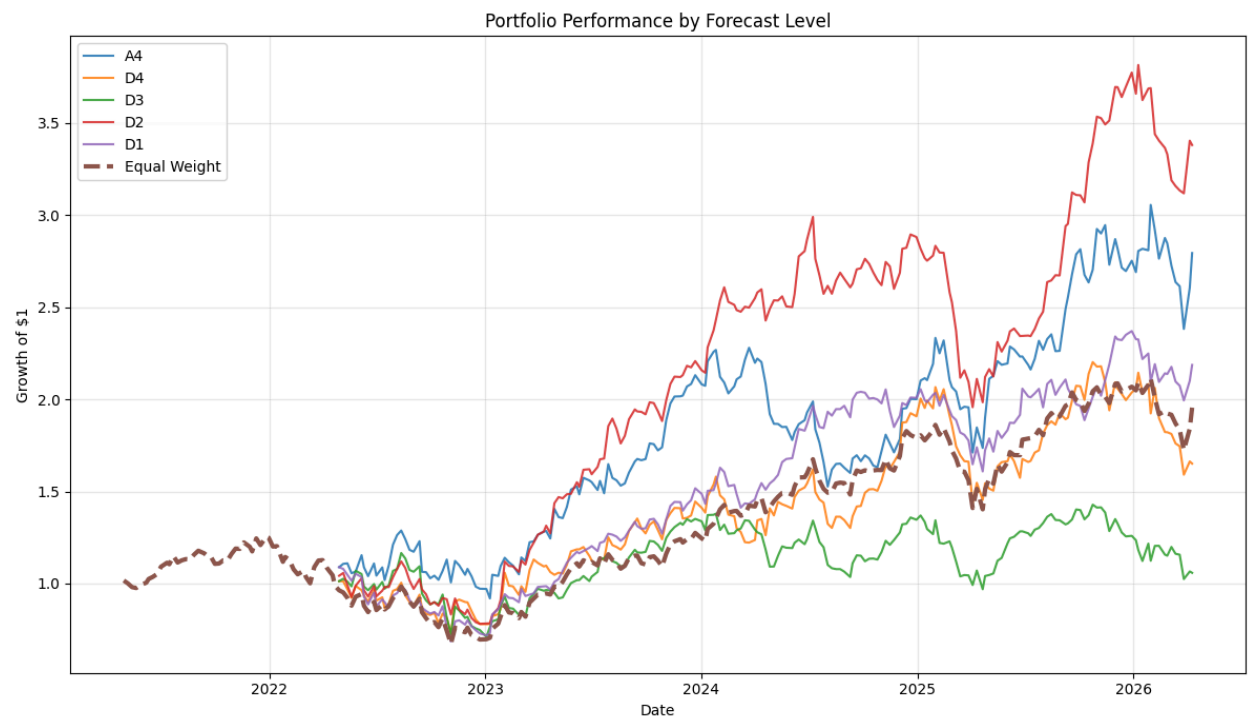
Level comparison summary:

	mean_period_simple_return	total_simple_return \	
A4	0.006060	1.794191	
D4	0.003512	0.623270	
D3	0.001860	0.203665	
D2	0.006646	2.312838	
D1	0.004726	1.223667	
EqualWeight	0.003169	0.954888	

	annualized_simple_return	annualized_volatility	sharpe_like \
A4	0.294503	0.337751	0.933018
D4	0.129410	0.354642	0.514901
D3	0.047668	0.318825	0.303432
D2	0.351071	0.300224	1.151168
D1	0.227170	0.286909	0.856528
EqualWeight	0.144061	0.245952	0.669944

	final_cumulative_growth
A4	2.794191
D4	1.623270
D3	1.203665
D2	3.312838
D1	2.223667
Equal weight	1.954888

Daily forecast weekly rebalance



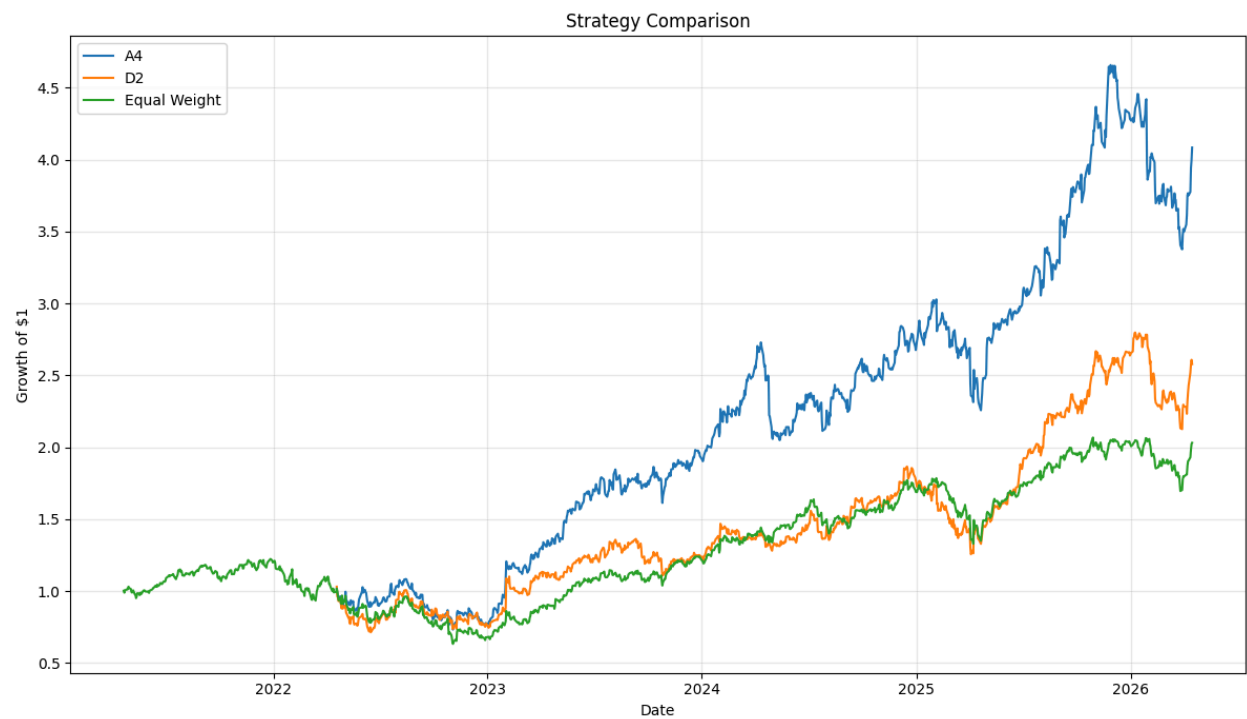
Level comparison summary:

	mean_period_simple_return	total_simple_return \		
A4	0.006060	1.794265		
D4	0.003601	0.651191		
D3	0.001223	0.058200		
D2	0.006739	2.380673		
D1	0.004635	1.186845		
EqualWeight	0.003169	0.954888		

	annualized_simple_return	annualized_volatility	sharpe_like \
A4	0.294511	0.337751	0.933038
D4	0.134259	0.355595	0.526532
D3	0.014312	0.315493	0.201643
D2	0.357969	0.299329	1.170787
D1	0.221932	0.285053	0.845499
EqualWeight	0.144061	0.245952	0.669943

	final_cumulative_growth
A4	2.794265
D4	1.651191
D3	1.058200
D2	3.380673
D1	2.186845
EqualWeight	1.954888

Daily forecast daily rebalance



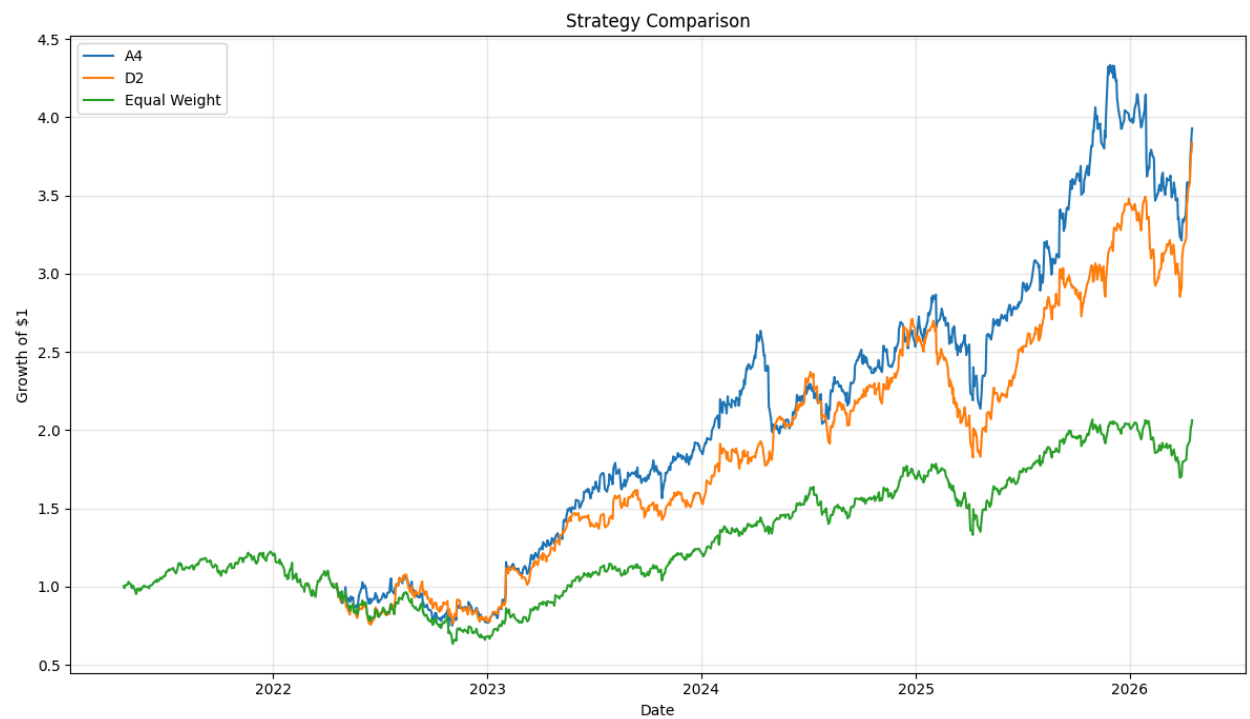
Strategy Summary:

	mean_period_simple_return	total_simple_return \		
A4	0.001642	3.083968		
D2	0.001174	1.578405		
Equal Weight	0.000707	1.031247		

	annualized_simple_return	annualized_volatility	sharpe_like \	
A4	0.425584	0.346299	1.195162	
D2	0.268976	0.342782	0.863001	
Equal Weight	0.153178	0.267099	0.667050	

	final_cumulative_growth	
A4	4.083968	
D2	2.578405	
Equal Weight	2.031247	

A4 vs D1 (misabeled)



Strategy Summary:

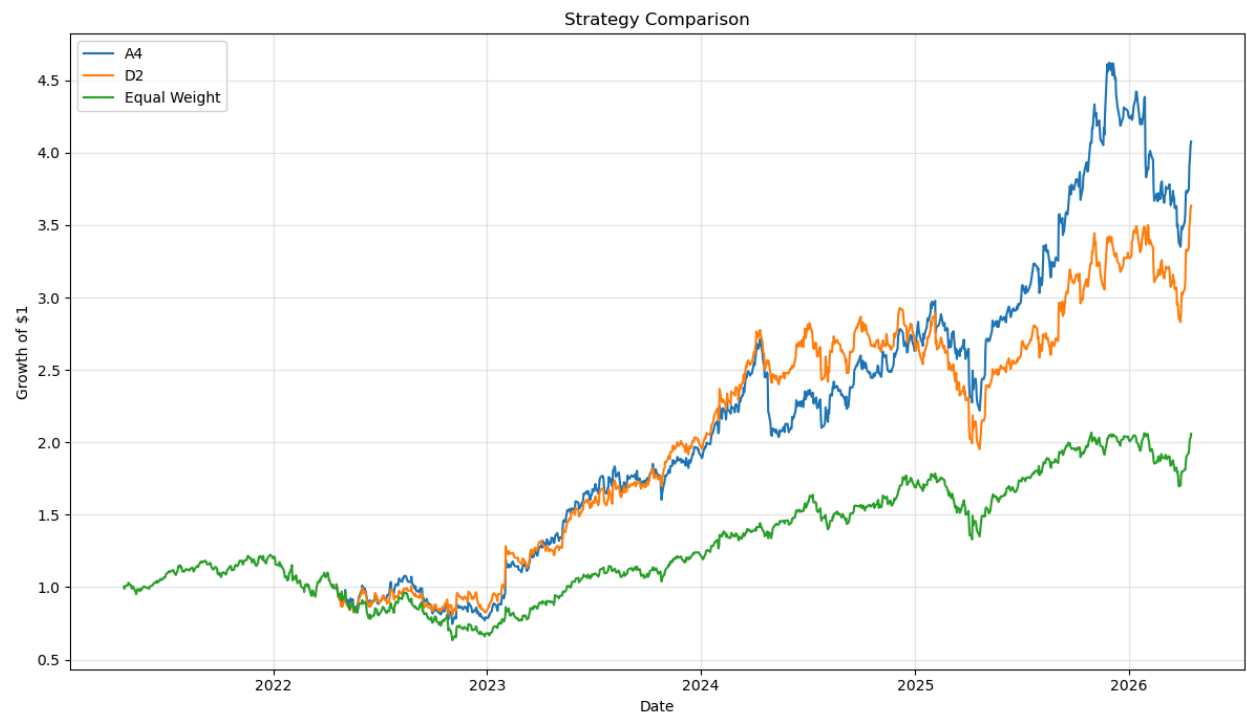
	mean_period_simple_return	total_simple_return \		
A4	0.001610	2.928940		
D1	0.001602	2.836043		
Equal Weight	0.000719	1.063176		

	annualized_simple_return	annualized_volatility	sharpe_like \	
A4	0.414197	0.346018	1.172722	
D1	0.413497	0.342857	1.177672	
Equal Weight	0.156667	0.267077	0.678400	

	final_cumulative_growth	
A4	3.928940	
D1	3.836043	
Equal Weight	2.063176	

Average deviation from equal weight:

A4 vs A4 + D1



Strategy Summary:

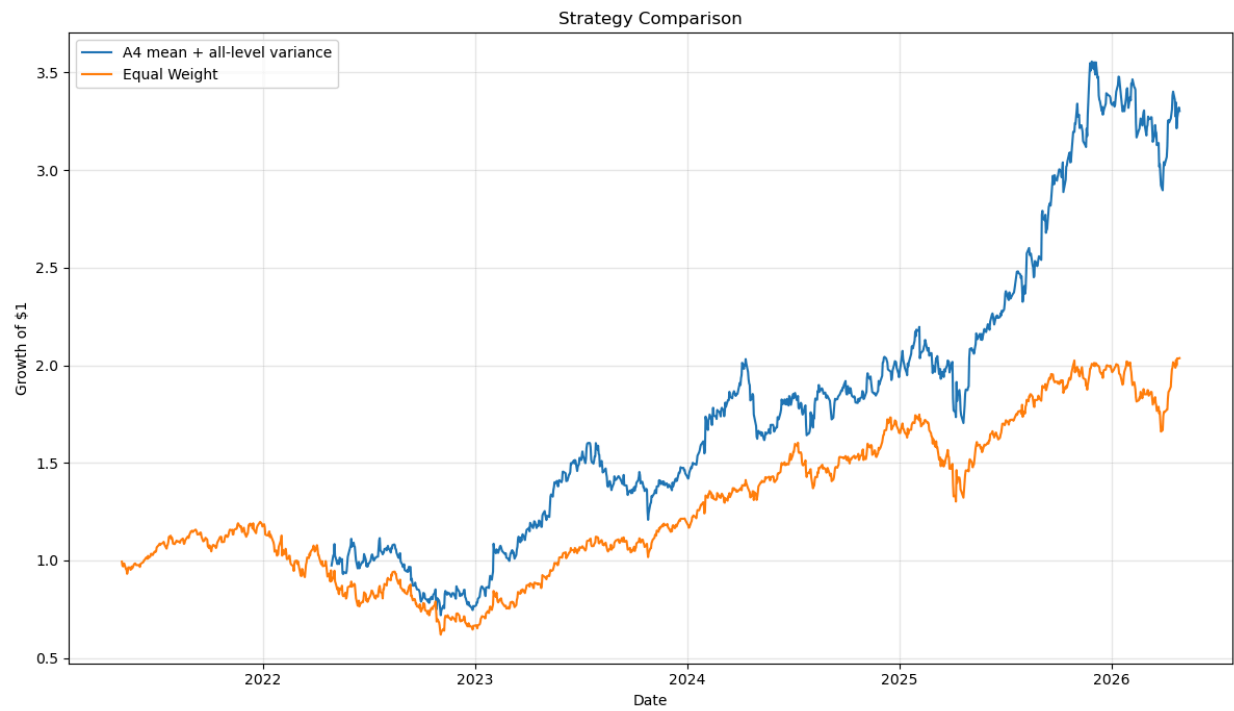
	mean_period_simple_return	total_simple_return \		
A4	0.001643	3.077437		
D2	0.001554	2.634006		
Equal Weight	0.000718	1.059411		

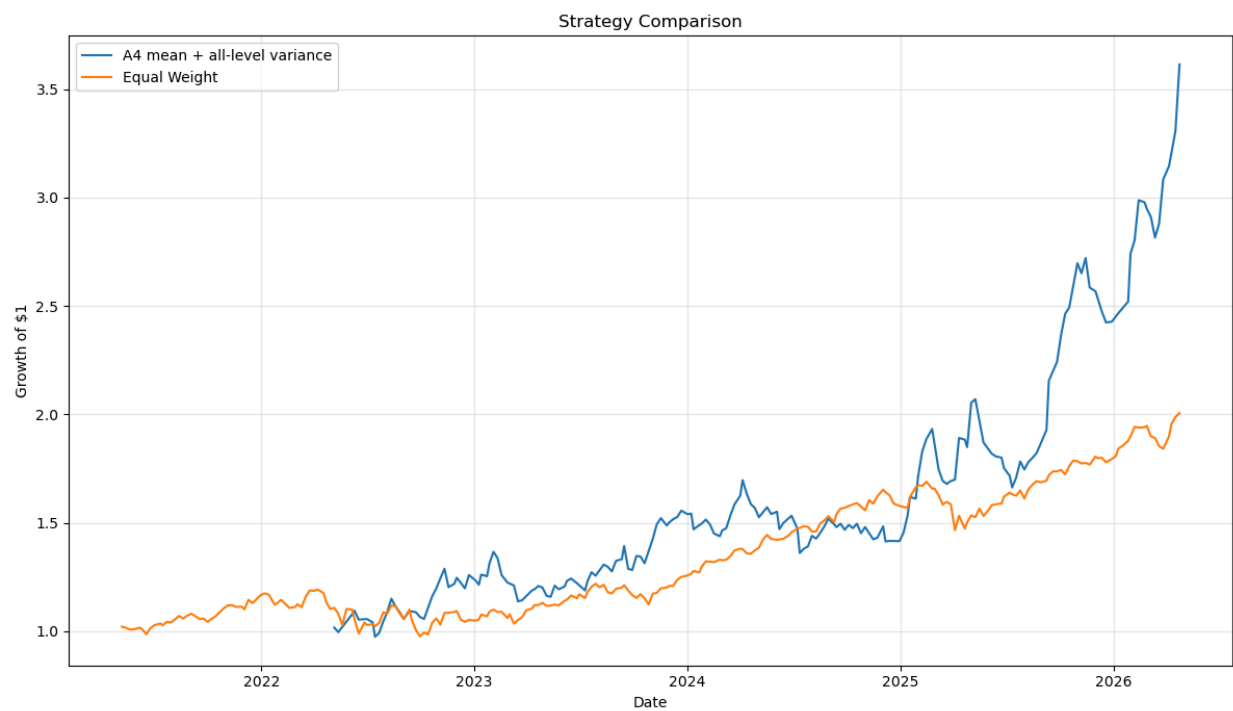
	annualized_simple_return	annualized_volatility	sharpe_like \	
A4	0.426021	0.345721	1.197503	
D2	0.396805	0.341691	1.146089	
Equal Weight	0.156242	0.267058	0.677056	

	final_cumulative_growth	
A4	4.077437	
D2	3.634006	
Equal Weight	2.059411	

A4 mean + all-level variance summary:

	mean_period_simple_return	total_simple_return	annualized_simple_return
annualized_volatility	sharpe_like	final_cumulative_growth	
A4 mean + all-level variance	0.001440	2.303132	0.362693
0.327471	1.107924	3.303132	
Equal Weight	0.000709	1.036676	0.153797
0.267213	0.668888	2.036676	





A4 mean + all-level variance summary:

	mean_period_simple_return	total_simple_return	annualized_simple_return
A4 mean + all-level variance	0.007654	2.613692	0.434926
0.269155	1.478705	3.613692	
Equal Weight	0.002866	1.005317	0.149308
0.139549	1.068139	2.005317	

Progress: 0.4% (1/261)
Progress: 4.2% (11/261)
Progress: 8.0% (21/261)
Progress: 11.9% (31/261)
Progress: 15.7% (41/261)
Progress: 19.5% (51/261)
Progress: 23.4% (61/261)
Progress: 27.2% (71/261)
Progress: 31.0% (81/261)
Progress: 34.9% (91/261)
Progress: 38.7% (101/261)
Progress: 42.5% (111/261)
Progress: 46.4% (121/261)
Progress: 50.2% (131/261)
Progress: 54.0% (141/261)
Progress: 57.9% (151/261)
Progress: 61.7% (161/261)
Progress: 65.5% (171/261)
Progress: 69.3% (181/261)
Progress: 73.2% (191/261)
Progress: 77.0% (201/261)
Progress: 80.8% (211/261)
Progress: 84.7% (221/261)
Progress: 88.5% (231/261)
Progress: 92.3% (241/261)
Progress: 96.2% (251/261)
Progress: 100.0% (261/261)
Progress: 0.4% (1/261)
Progress: 4.2% (11/261)
Progress: 8.0% (21/261)
Progress: 11.9% (31/261)
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Progress: 69.3% (181/261)
Progress: 73.2% (191/261)
Progress: 77.0% (201/261)
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Progress: 88.5% (231/261)
Progress: 92.3% (241/261)
Progress: 96.2% (251/261)
Progress: 100.0% (261/261)
Progress: 0.4% (1/261)
Progress: 4.2% (11/261)
Progress: 8.0% (21/261)
Progress: 11.9% (31/261)
Progress: 15.7% (41/261)
Progress: 19.5% (51/261)
Progress: 23.4% (61/261)
Progress: 27.2% (71/261)

Skipping 2022-12-02 because: ARMA-GARCH optimization failed: ABNORMAL:

Progress: 31.0% (81/261)
Progress: 34.9% (91/261)
Progress: 38.7% (101/261)
Progress: 42.5% (111/261)
Progress: 46.4% (121/261)
Progress: 50.2% (131/261)
Progress: 54.0% (141/261)
Progress: 57.9% (151/261)
Progress: 61.7% (161/261)
Progress: 65.5% (171/261)
Progress: 69.3% (181/261)
Progress: 73.2% (191/261)
Progress: 77.0% (201/261)
Progress: 80.8% (211/261)
Progress: 84.7% (221/261)
Progress: 88.5% (231/261)
Progress: 92.3% (241/261)
Progress: 96.2% (251/261)
Progress: 100.0% (261/261)
Progress: 0.4% (1/261)
Progress: 4.2% (11/261)
Progress: 8.0% (21/261)
Progress: 11.9% (31/261)
Progress: 15.7% (41/261)

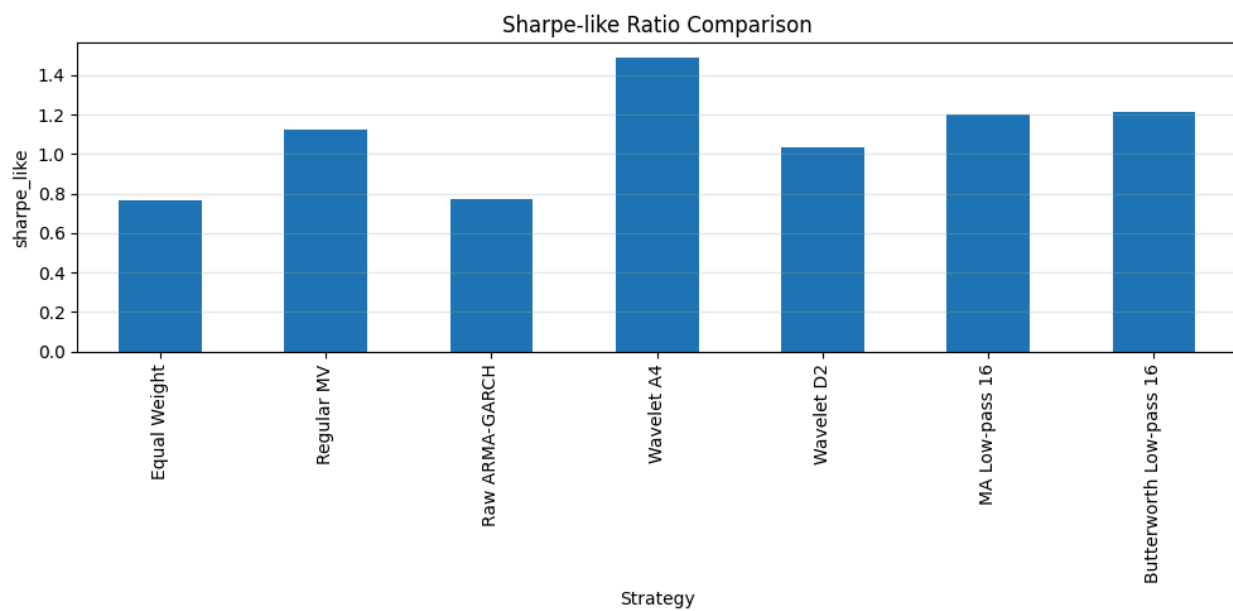
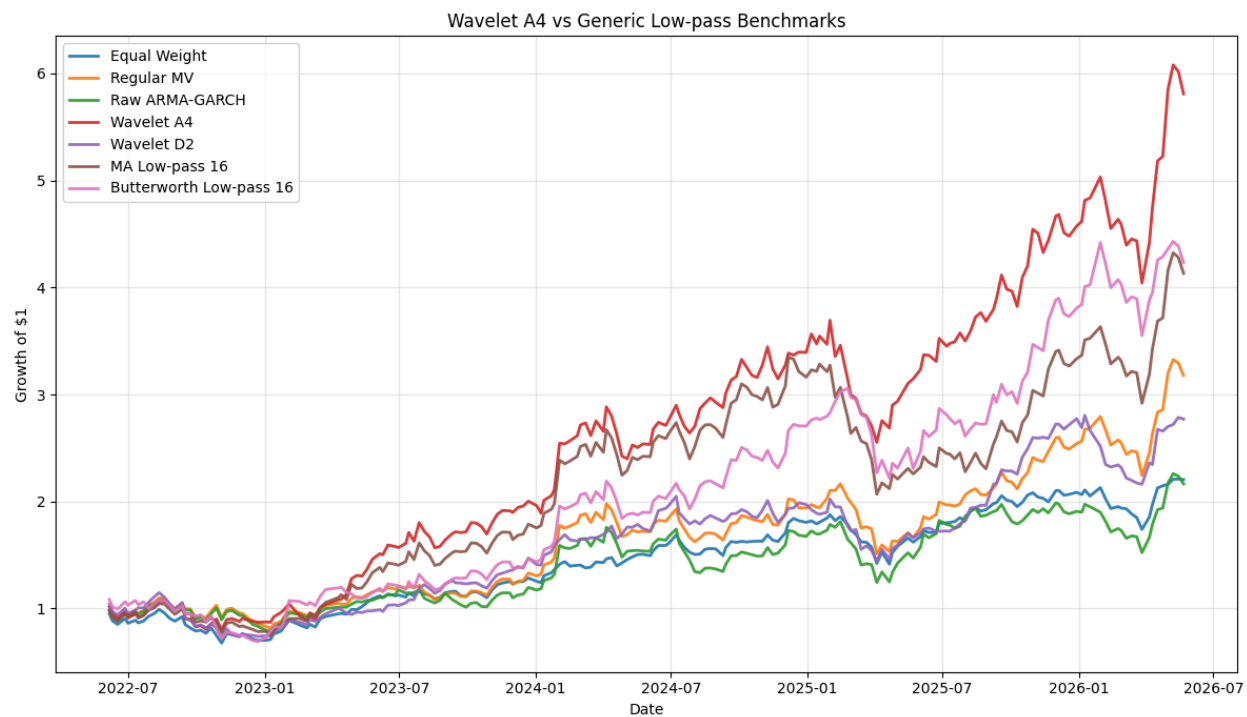
Progress: 19.5% (51/261)
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Progress: 27.2% (71/261)
Progress: 31.0% (81/261)
Progress: 34.9% (91/261)
Progress: 38.7% (101/261)
Progress: 42.5% (111/261)
Progress: 46.4% (121/261)
Progress: 50.2% (131/261)
Progress: 54.0% (141/261)
Progress: 57.9% (151/261)
Progress: 61.7% (161/261)
Progress: 65.5% (171/261)
Progress: 69.3% (181/261)
Progress: 73.2% (191/261)
Progress: 77.0% (201/261)
Progress: 80.8% (211/261)
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Progress: 88.5% (231/261)
Progress: 92.3% (241/261)
Progress: 96.2% (251/261)
Progress: 100.0% (261/261)
Progress: 0.4% (1/261)
Progress: 4.2% (11/261)
Progress: 8.0% (21/261)
Progress: 11.9% (31/261)
Progress: 15.7% (41/261)
Progress: 19.5% (51/261)
Progress: 23.4% (61/261)
Progress: 27.2% (71/261)
Progress: 31.0% (81/261)
Progress: 34.9% (91/261)
Progress: 38.7% (101/261)
Progress: 42.5% (111/261)
Progress: 46.4% (121/261)
Progress: 50.2% (131/261)
Progress: 54.0% (141/261)
Progress: 57.9% (151/261)
Progress: 61.7% (161/261)
Progress: 65.5% (171/261)
Progress: 69.3% (181/261)
Progress: 73.2% (191/261)
Progress: 77.0% (201/261)
Progress: 80.8% (211/261)

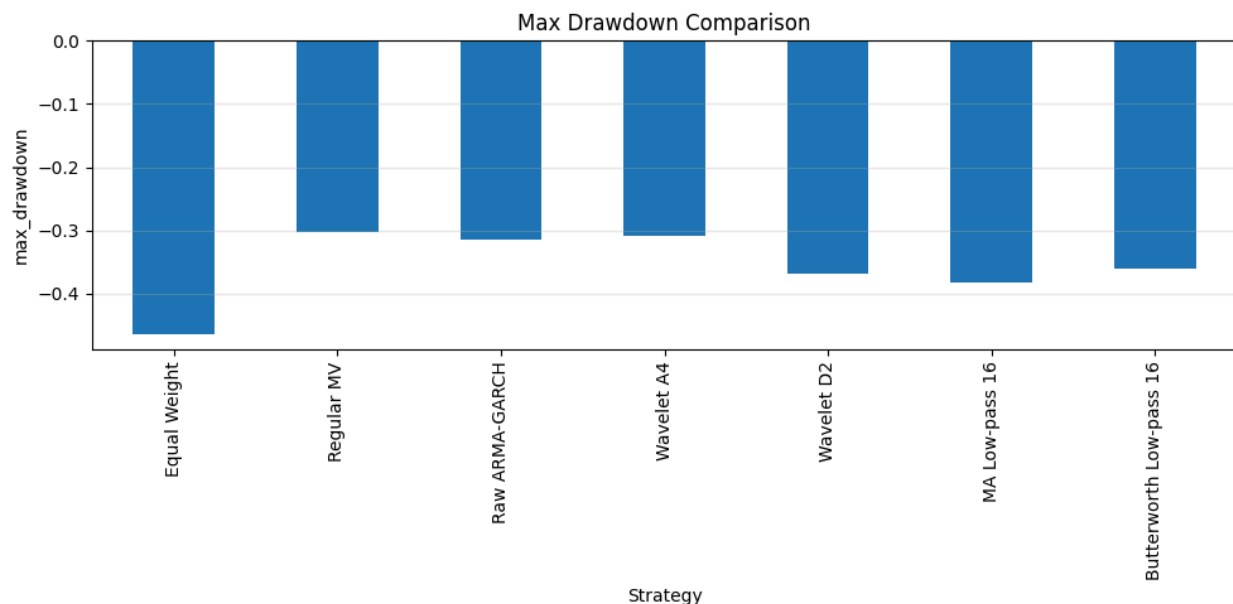
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 Progress: 96.2% (251/261)
 Progress: 100.0% (261/261)
 Progress: 0.4% (1/261)
 Progress: 4.2% (11/261)
 Progress: 8.0% (21/261)
 Progress: 11.9% (31/261)
 Progress: 15.7% (41/261)
 Progress: 19.5% (51/261)
 Progress: 23.4% (61/261)
 Progress: 27.2% (71/261)
 Progress: 31.0% (81/261)
 Skipping 2023-02-03 because: ARMA-GARCH optimization failed: ABNORMAL:
 Progress: 34.9% (91/261)
 Skipping 2023-03-24 because: ARMA-GARCH optimization failed: ABNORMAL:
 Progress: 38.7% (101/261)
 Progress: 42.5% (111/261)
 Progress: 46.4% (121/261)
 Progress: 50.2% (131/261)
 Progress: 54.0% (141/261)
 Progress: 57.9% (151/261)
 Progress: 61.7% (161/261)
 Progress: 65.5% (171/261)
 Progress: 69.3% (181/261)
 Progress: 73.2% (191/261)
 Progress: 77.0% (201/261)
 Skipping 2025-05-02 because: ARMA-GARCH optimization failed: ABNORMAL:
 Progress: 80.8% (211/261)
 Progress: 84.7% (221/261)
 Progress: 88.5% (231/261)
 Progress: 92.3% (241/261)
 Progress: 96.2% (251/261)
 Progress: 100.0% (261/261)

	n	mean_	median	best_	worst_	w	total	annu	ann	s	ma	c	final_
	_	period	_perio	period	period	i	_si	alized	ualiz	h	x_	al	cumu
	p	_simpl	d_simpl	_simpl	_simpl	_	mpl	_simp	ed_	ar	dra	m	lative
	e	e_retu	d_simp	_simp	_simp	r	e_re	le_ret	vola	p	wd	ar	_gro
	ri	rn	le_retu	le_ret	e_retu	a	turn	urn	tility	_l	ow	_l	wth
	o		rn	urn	rn						n		

	d s					t e				ik e		ik e	
E q u a l W e i g h t	2 6 0 . 0	0.0036 29	0.0026 74	0.1314 49	-0.096 627	0 . 5 5 3 8 4 6	1.20 480 5	0.171 316	0.24 7438	0. 7 6 2 7 4 9	-0. 46 41 83	0. 3 6 9 0 7 0	2.204 805
R e g u l a r M V	2 0 8 . 0	0.0064 09	0.0038 30	0.2051 30	-0.133 933	0 . 5 2 4 0 3 8	2.17 796 8	0.335 173	0.29 6928	1. 1 2 2 3 7 9	-0. 30 21 24	1. 1 0 9 3 8 7	3.177 968
R a w A R M A- G A R C H	2 0 8 . 0	0.0046 46	0.0015 24	0.2051 30	-0.118 269	0 . 5 2 4 0 3 8	1.16 457 3	0.212 950	0.31 2687	0. 7 7 2 6 3 6	-0. 31 38 73	0. 6 7 8 4 6 1	2.164 573
W a v e l e t A 4	2 0 7 . 0	0.0095 99	0.0096 65	0.2051 30	-0.135 146	0 . 5 6 5 2 1 7	4.80 884 5	0.555 770	0.33 5443	1. 4 8 8 0 2 0	-0. 30 87 38	1. 8 0 1 3 5	5.808 845

W a v e l e t D 2	2 0 8 . 0	0.0057 00	0.0038 02	0.1399 95	-0.135 146	0 . 5 4 8 0 7 7	1.76 995 7	0.290 085	0.28 7277	1. 0 3 1 7 4 0	-0. 36 74 22	0. 7 8 9 5 1 4	2.769 957
M A L o w -p a s s 1 6	2 0 7 . 0	0.0080 11	0.0072 67	0.2051 30	-0.135 475	0 . 5 3 6 2 3 2	3.13 151 3	0.428 142	0.34 6646	1. 2 0 1 6 9 4	-0. 38 25 02	1. 1 1 9 3 2 0	4.131 513
B u t t e r w o r t h L o w -p a s s 1 6	2 0 5 . 0	0.0082 23	0.0043 52	0.2292 73	-0.135 475	0 . 5 2 6 8 2 9	3.23 352 9	0.442 011	0.35 1669	1. 2 1 5 8 7 6	-0. 36 04 22	1. 2 2 6 3 6 9	4.233 529





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